Why Some Recordings Get Rejected

A plain-English guide to the quarantine policy

Every recording in this dataset is supposed to have the same seven columns, in the same order: `Time, Vib_Spindle, Vib_Table, X_Load_Cell, Y_Load_Cell, Z_Load_Cell, Current`. That shared layout is what makes it possible to compare recordings, build a baseline for each condition, and cite an exact evidence window when the assistant answers a question. This shared layout is called the **canonical schema**.

One structural detail worth knowing: a canonical file actually has **two** header-like rows before the numeric data starts — the column-name row, then a units row reading `Unit,V,V,V,V,V,V`. A parser that only skips one header row would try to read that units row as data and choke on the word “V” where it expects a number. Confirming a file's quality correctly means recognising and skipping both rows, not just the first.

What happens when a file doesn't match

A small number of files don't have all seven columns, don't have a readable header row, or contain values that aren't valid numbers. When that happens, the system does **not** try to guess, repair, or force the file into the expected shape. Instead it sets that file aside — **quarantines** it — and writes down exactly why.

This is a deliberate data-integrity choice, not a bug. "Fixing" a file with a missing column by guessing which channel is missing would mean inventing a Z-axis load reading, or a Current reading, that was never actually measured. An answer built on an invented number is worse than no answer at all — it looks trustworthy but isn't. Quarantine keeps that from ever happening silently.

The current numbers

Discovered	Canonical (used)	Quarantined (set aside)
822 recordings	805 recordings	17 recordings

That's roughly 98% of the corpus passing the check. The 17 quarantined files are still on disk, untouched — nothing is deleted — and each one has a logged reason.

The kinds of reasons a file gets quarantined

Reason	In plain English
Missing or wrong header	The first row doesn't say Time, Vib_Spindle, etc. in the expected order, so the system can't be sure which column is which.
Wrong number of columns	The file has more or fewer than seven columns — a sensor may have failed to log, or the file may be from a different setup entirely.
Non-numeric values	A value that should be a number (a vibration reading, a force reading) contains text or is unreadable.
File couldn't be parsed	The file is corrupted, unusually large, or otherwise unreadable by the standard CSV reader.

One more category is worth knowing about: some files contain a few stray invisible characters (called null bytes) left over from how they were originally written. Those characters are automatically cleaned out and the cleaning is logged — that file still gets used normally. That's different from quarantine: it's a small, logged repair to the file's encoding, never to its actual sensor data.

What this means for you

When you ask a question, the answer is always built only from the 805 canonical recordings. If you ever wonder whether a specific recording was used, you can check the Dataset Health page or the quarantine log — every rejected file and its reason is there, not hidden.

Source: README.md §2 ("Ingestion and quality gate"), config/domain.yaml quality_policy, and the project's quarantine log design.